Chromostereopsis pilot analysis

▼ Background

The chromostereopsis effect varies between observers. With red and blue on a dark background, most people see red as closer (red-near observers), a minority see blue as closer (blue-near observers) and some see no depth difference, which largely reflects individual differences in pupil position and eye optics (Kitaoka, 2015). Since chromostereopsis affects the apparent depth of red and blue surfaces, it may also bias their perceived size.

Two accounts, size constancy and irradiation, predict opposite size biases [the project overview is in IllusionGameChromostereopsis > AGENTS.md]. Under size constancy, the bias follows perceived depth and the nearer disc is judged as smaller. In this case, red would be judged as smaller by red-near observers, as larger by blue-near observers, and not biased by those who see no depth. Under the irradiation account, the red-leaning disc, which is typically brighter, would be judged as larger regardless of perceived depth, and if luminance was accounted for, there would be no bias on size perception.

▼ Pilot aim

The pilot was designed as an initial check of whether chromostereopsis biases size judgements of red-blue discs, in which direction (as predicted by size constancy or irradiation accounts), and how stimulus settings modulate the effect.

▼ Pilot analysis summary

- Pilot data: 15 participants, 3,750 trials (250 per participant)
- Tables and figures are in IllusionGameChromostereopsis > analysis > analysis.html
- The pilot provides preliminary, suggestive evidence of a small effect in the direction predicted by the size-constancy account, with a slight tendency for red and red-leaning discs to be judged as smaller. However, the evidence remains inconclusive and the effect estimate is uncertain given the small participant sample (n = 15). The observed variability across participants suggests that individual differences may be important, although their magnitude is uncertain. The pilot also provides provisional guidance for stimulus selection, although the optimal settings have not been established.

▼ Evidence for the illusion

- Effect estimates are exploratory and should be interpreted cautiously given the small participant sample (n = 15). Statistical tests are reported as supplementary evidence rather than treated as definitive evidence for or against an effect.
- The models predict the probability of a right-arrow response (the participant said the right disc was larger). Illusion strength is signed by the side of the red-leaning disc (positive = left, negative = right, 0 = no red/blue difference). A positive coefficient therefore means the red-leaning disc tended to be judged as smaller: with it on the left, more right-arrow responses means the left disc is judged smaller, and with it on the right, more left-arrow responses means the right disc is judged smaller.
- Core model: b = 0.42 log-odds per unit of illusion strength, likelihood-ratio test p = .054 (Wald p = .040). Bootstrap 95% CI [0.05, 0.82]. The difference between the tests highlights the uncertainty of the inference.

- Bayesian model: b = 0.41, 95% CrI [-0.08, 0.89], with a 96% posterior probability that the effect is positive. The credible interval includes zero, which indicates substantial uncertainty about the effect's magnitude and direction.

▼ **Individual differences**

- Approximately 10 of 15 participants had positive estimated effects, indicating a tendency to judge the red-leaning disc as smaller.
- Approximately 3 participants had estimated effects near zero, suggesting little clear bias in either direction.
- Approximately 2 participants had negative estimated effects, indicating a tendency to judge the red-leaning disc as larger.
- These estimates are exploratory and should not be interpreted as establishing distinct participant subgroups or stable individual differences.

▼ **Parameter choices**

- Dither size was the only setting for which there was preliminary evidence of a difference in the estimated effect. The estimated effect was smaller at 2 px than at 4 px and 8 px, but this finding did not remain significant after correction for multiple comparisons. A dither size of 4 px or 8 px is therefore a provisional choice rather than an established optimum.
- No other setting showed reliable evidence of moderating the effect. Choices can instead be guided by practical considerations and the aim to reduce potential confounds.

▼ **Interpretation in relation to the literature**

- The direction of the estimated effect appeared to be in line with the size constancy prediction, though the evidence remains inconclusive.
- The estimated effect had the same direction in trials that were not equiluminant (0.09, 95% CI [-0.01, 0.19]), where the red-leaning disc is brighter and irradiation predicts it would be judged as larger. However, this estimate is descriptive.
- The variability in estimated effects across participants is compatible with known individual differences in chromostereopsis (Kitaoka, 2015). However, perceived depth was not measured, so this explanation could not be tested directly.
- Note that the evidence was weak, and interpretation is limited by the small sample.

▼ Descriptive accuracy results

Pooled accuracy across all trials:

- No red/blue difference (baseline, illusion strength = 0): 80% correct
- Red or red-leaning disc was actually the smaller disc: 83% correct
- Red or red-leaning disc was actually the larger disc: 74% correct

Accuracy within participants, with a difference metric (accuracy when the red-leaning disc was actually smaller minus accuracy when it was actually larger; a positive difference means higher accuracy when the red-leaning disc was actually smaller):

- No clear effect either way (6 participants): P03, P07, P08, P10, P11 and P15 (difference within ±0.05)
- More accurate when the red-leaning disc was actually smaller (8 participants): P02, P04, P06, P09, P12, P13, P14 and P16 (difference > 0.05)

- More accurate when the red-leaning disc was actually larger (1 participant): P17 at −0.20 (difference < −0.05)

NB: These comparisons are descriptive; no statistical tests were run. Participants were grouped using an arbitrary cut-off (±0.05) on the accuracy difference.

▼ Core model results

▼ Fixed effects

```
# Fixed Effects

Parameter             | Log-Odds |   SE |         95% CI |     z |      p
-------------------------------------------------------------------------
(Intercept)           |    -0.10 | 0.11 | [-0.33,  0.12] | -0.88 | 0.377
Area Difference 10pct |    -1.17 | 0.14 | [-1.44, -0.89] | -8.24 | < .001
Illusion Strength     |     0.42 | 0.20 | [ 0.02,  0.82] |  2.05 | 0.040
```

- Intercept: no clear evidence of an overall left/right bias
- Area difference is significant implies participants detect real size differences
- Illusion strength (0.42, p = .04): a small positive effect on average, where the red or red-leaning disc was judged as slightly smaller (in line with the size-constancy prediction). Participants varied around this average (see random effects)
- Small sample size caution, especially for p-value (Wald p-values optimistic with few participants)

▼ Random effects

```
# Random Effects

Parameter                                                | Coefficient
-----------------------------------------------------------------------
SD (Intercept: Participant)                              |        0.41
SD (Area_Difference_10pct: Participant)                  |        0.51
SD (Illusion_Strength: Participant)                      |        0.75
Cor (Intercept~Area_Difference_10pct: Participant)       |        0.19
Cor (Intercept~Illusion_Strength: Participant)           |       -0.25
Cor (Area_Difference_10pct~Illusion_Strength: Participant) |        0.28
```

- Participants differed in their:
  - baseline response bias (Intercept SD = 0.41), meaning the bias towards choosing left/right arrow regardless of stimulus presentation varied across participants
  - size discrimination ability (Difference SD = 0.51), meaning the difference in the ability of participants to detect real size differences, outside of the effect of the illusion strength (red/blue contrast) varied
  - colour illusion sensitivity (Illusion Strength SD = 0.75), meaning the degree to which illusion strength biased the response choice varied across participants. This SD exceeded the average estimated illusion effect (0.42)

▼ Likelihood-ratio test (p = .054)

```
Data: df
Models:
m_null: ResponseRight ~ Area_Difference_10pct + (1 + Area_Difference_10pct + Illusion_Str
m_core: ResponseRight ~ Area_Difference_10pct + Illusion_Strength + (1 + Area_Difference_
       npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
m_null    8 3249.0 3298.8 -1616.5    3233.0
m_core    9 3247.3 3303.3 -1614.6    3229.3 3.718  1    0.05383 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- The likelihood-ratio test was non-significant ($\chi^2(1) = 3.72$, $p = .054$), so there is insufficient evidence at $\alpha = .05$ to suggest that illusion strength affects responses

▼ Bootstrapping for confidence intervals

Bootstrap warnings (2.4% of simulations)

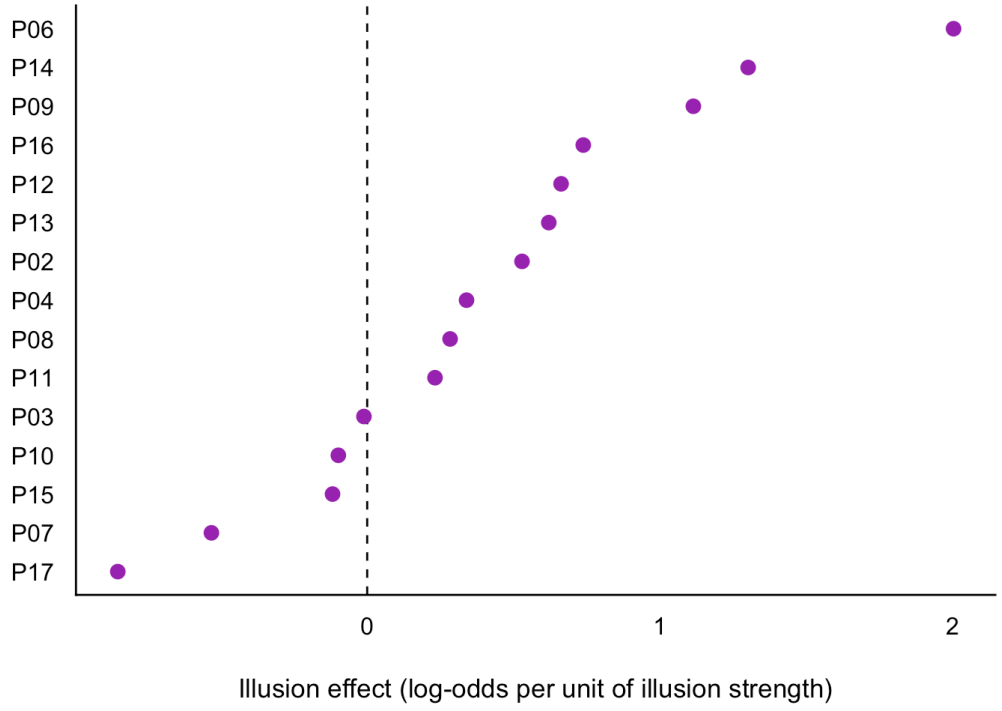
- singular fit (10/500): Some simulated datasets produced models where one or more participant-level variation estimates were near zero
- convergence warnings (2/500): Some models had difficulty finding a stable solution

Confidence intervals for parameter estimates

```
                              2.5 %     97.5 %
(Intercept)           -0.35468423  0.1184665
Area_Difference_10pct -1.45758999 -0.9082146
Illusion_Strength      0.04796233  0.8189996
```

- The intercept interval includes 0 which indicates no clear baseline bias towards left or right arrow pressing
- The difference interval excludes 0 which indicates evidence of size discrimination
- The illusion strength interval just excludes zero which indicates the possibility of a small positive effect
- However, the likelihood-ratio test gave $p = .054$, so the overall evidence remains borderline

▼ Individual differences



- Red or red-leaning disc judged smaller (10 participants): P06 (about 2), P14 (about 1.3) and P09 (about 1.1). Then P16, P12, P13, P02, P04, P08 and P11 range from about 0.7 down to 0.2
- Essentially no effect (3 participants): P03, P10 and P15 sit within about 0.1 of zero
- Reversed, red or red-leaning disc judged larger (2 participants): P07 (about -0.5) and P17 (about -0.8) but with only 250 trials be cautious

▼ Linear effect of illusion strength?

```
Data: df
Models:
m_core: ResponseRight ~ Area_Difference_10pct + Illusion_Strength + (1 + Area_Difference_
m_cat: ResponseRight ~ Area_Difference_10pct + (1 + Area_Difference_10pct + Illusion_Stre
       npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
m_core    9 3247.3 3303.3 -1614.6    3229.3
m_cat    12 3253.2 3328.0 -1614.6    3229.2 0.0322  3     0.9985
```

- The core model treats illusion strength as a continuous predictor, to test this assumption, it is compared against a model that treats illusion strength as a categorical factor (-1, -0.5, 0, 0.5, 1)
- The linear model fit as well as the categorical model ($\chi^2(3) = 0.03$, $p = .99$)
- Both AIC (3247.3 vs 3253.2) and BIC (3303.3 vs 3328.0) favoured the more parsimonious linear model

  → no changes to the model

▼ Stimulus settings

Factor	Chisq	df	p	p_holm	p_bonferroni	p_bh
Size_f	3.14	2	0.21	1.00	1.00	0.62
Gap_f	0.88	2	0.64	1.00	1.00	0.77
Density_f	0.38	2	0.83	1.00	1.00	0.83
Dither_Size_f	8.98	2	0.01	0.07	0.07	0.07
Equiluminant	0.31	1	0.58	1.00	1.00	0.77
Size_Panel_Mode	0.84	1	0.36	1.00	1.00	0.72

```
Dither_Size_f | Slope |   SE |        95% CI |    z |     p
-------------------------------------------------------------
2             |  0.03 | 0.05 | [-0.07, 0.14] | 0.64 | 0.523
4             |  0.13 | 0.05 | [ 0.03, 0.24] | 2.46 | 0.014
8             |  0.13 | 0.05 | [ 0.03, 0.24] | 2.48 | 0.013

Marginal effects estimated for Illusion_Strength
Type of slope was dY/dXEstimated Marginal Effects
```

Dither

- Only dither size had evidence of moderation, and this effect did not sustain correction for multiple comparisons
- the illusion was weak at the finest dither, 2 px (0.03, p = .5) and similar at 4 px and 8 px (0.13 with p = .01 for each)
- note that about a third of trials at each dither size used density 1 (solid fill), where dither size has no visible effect, which may dilute any dither contrast

Others

- No other moderator showed reliable differences at this sample size
- Remaining parameters can therefore be set for practical or interpretive reasons (e.g. equiluminant stimuli to rule out brightness confounds) rather than to maximise the effect

▼ Recommended settings

No setting choice is specifically supported after correction for multiple comparisons. The choices below follow the point estimates (and, for equiluminance, a design rationale), and should be treated as provisional defaults.

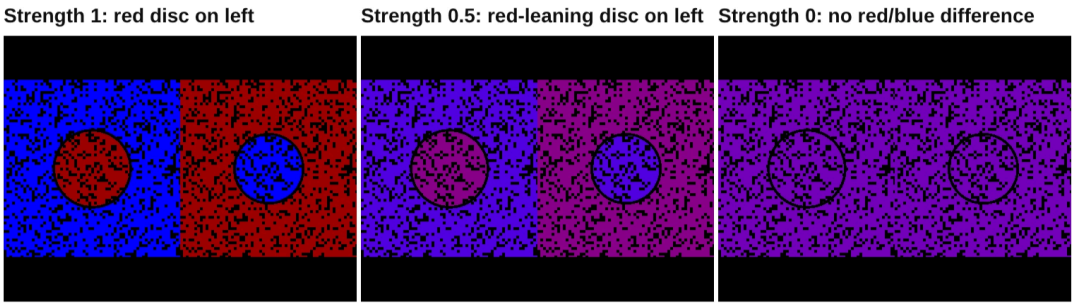
- Equiluminant = TRUE. Estimates were similar (0.11 vs 0.09; interaction p = .58) but it would remove the brightness difference between the discs as a potential confound.
- Dither size = 4 px or 8 px. The estimated contribution was 0.13 at both 4 px and 8 px, against 0.03 at 2 px.
- Gap = 0.02. The estimate fell descriptively as the gap widened (0.12, 0.10, 0.08), but not significantly (p = .64). Gap = 0 would remove the outline, but strength 0 with gap 0 was excluded in the pilot (the disc would be invisible), so including a baseline condition needs a gap above 0. 0.02 is the smallest gap used and was close to the gap 0 estimate.
- Size = Any. The estimate was highest descriptively at the smallest size (0.14 vs 0.08 at both larger sizes), but not significantly different (p = .21). Choose for what renders best on both a desktop and phone screen.

- Size panel mode = Any. ‘Fill’ may be the better design choice, as it reduces the extra comparison that can be made in the ‘size panel = match’ setting, where detecting the size within a square may contribute to decision-making.
- Density = Any. No clear pattern. Note that density = 1 is a solid fill with no visible dither, so density should be below 1 if dither size is to apply (0.75 in the example below).

NB: These settings were chosen from separate tests and have not been tested in combination.

Example recommended settings:

Equiluminant TRUE, dither size 4 px, gap 0.02, size 0.5, size panel fill, density 0.75. Left disc larger. Only illusion strength differs between panels.



▼ Bayesian model

```
Summary of Posterior Distribution

Parameter             | Median |         95% CI |     pd |  Rhat | ESS (tail)
-------------------------------------------------------------------------------
(Intercept)           |  -0.10 | [-0.37,  0.19] | 77.15% | 1.004 |       1804
Area_Difference_10pct |  -1.17 | [-1.51, -0.84] |   100% | 1.002 |       2046
Illusion_Strength     |   0.41 | [-0.08,  0.89] | 95.62% | 1.001 |       2004
```

- Bayesian version of the core model gave a consistent estimate (b = 0.41, 95% CrI [-0.08, 0.89]) with a 95.6% posterior probability that the illusion was positive
- The credible interval was slightly wider than the frequentist ones and included zero
- Sensitivity checks on the prior were not conducted

▼ Still to check

- Viewing conditions and eyesight: screen size, pixel density, device and input type, and colour vision
- Task difficulty: accuracy by difficulty level, and whether the easiest levels are near ceiling and so uninformative
- Order effects: practice, fatigue or strategy changes across blocks
- Setting combination: whether the chosen settings together give at least the effect seen here

▼ References

Kitaoka, A. (2015). Chromostereopsis. In *Encyclopedia of Color Science and Technology*. Springer. https://doi.org/10.1007/978-3-642-27851-8_210-1